

SYSTEM SPECIFICATIONS

CAMPUS PRINT HUB

Technical Workflow & Device Management Blueprint

This document explains exactly how the Campus Print Hub system works end-to-end across the User Interface, Backend Server, Desktop Print Client, and the physical Printer Hardware.

1. The Frontend User Interface (Student Portal)

The frontend is a single-page React web application styled with Tailwind CSS.

Student Experience Flow:

1. **QR Code Scan:** The student scans a QR code posted at the print shop counter using their phone camera, which opens the portal

([https://\[subdomain\].trycloudflare.com](https://[subdomain].trycloudflare.com)).

2. **Form Input:** The student enters their **Name** and **Email** (for security logging and pickup verification).

3. **Document Upload:** The student drags & drops or clicks to select a document (PDF, PNG, JPG, DOC, or DOCX).

4. Print Settings Selection:

- **Print Mode:** Toggle between **Black & White** (default) or **Color**.
- **Copies:** Selects the number of copies needed (1 to 10).
- **Page Range Selection (Advanced):** Option to specify print ranges (e.g. `1-3, 5`) to print only a subset of the document.

5. **Send to Print:** The student clicks "Send to Print". The UI tracks the file upload progress bar in real-time from 0% to 100%.
6. **Print Token:** Once uploaded, a unique print token is displayed (e.g., `PRNT-270`) on the success card.
7. **Live Queue Panel:** The right-hand panel displays a live queue showing their job status (`Queued` → `Printing` with a live 0-100% progress bar → `Completed` / `Ready for Pickup`).

2. The Backend Server (Express + SSE)

The backend is an Express Node.js server running in the cloud (or locally on port 3001) that handles file ingestion, security sanitization, and real-time event streaming.

Key Technical Operations:

1. **File Ingestion:** Uses `multer` to accept the multipart file upload.
2. **Strict Security Sanitization:**
 - **File Size Capping:** Limits uploads to a maximum of 50MB inside Multer to prevent disk-exhaustion Denial of Service (DoS) attacks.
 - **Extension and MIME Match:** Cross-references browser-reported MIME types with the actual file extension.
 - **Magic Bytes Verification (Anti-Spoofing):** Inspects the first 4 bytes of the binary buffer to verify the file signature, blocking executable files renamed as PDFs (e.g., ``virus.exe`` renamed to ``assignment.pdf`` is blocked because it starts with executable headers ``MZ` / `4D 5A`` instead of ``%PDF` / `25 50 44 46``).
3. **Robust PDF Page Counting:** Instead of using regex on the raw buffer (which fails on encrypted, compressed, or complex nested page tree PDFs), the server loads the buffer using a pure JavaScript library `pdf-lib`:

```
import { PDFDocument } from 'pdf-lib';
const pdfDoc = await PDFDocument.load(pdfBuffer);
const pageCount = pdfDoc.getPageCount();
```

This guarantees 100% accuracy for all PDF documents.

4. **Database Registration:** Saves job details in `server/data/db.json` (stores Token, Student Name, File Path, Page Count, Copies, Print Mode, Page Range, and Status).

5. **Server-Sent Events (SSE) Stream:**

- Exposes a persistent connection endpoint `/api/jobs/stream`. The browser and the desktop client keep this connection open.
- When the file finishes uploading, the server broadcasts a `{ type: "new_job", job }` event over the SSE stream to all listeners.

6. **Disk Space Cleanup:** The moment a job's status changes to `completed`, the server automatically calls `fs.unlinkSync` on the file inside `server/uploads/` to delete the PDF/image. This keeps server storage clean.

3. The Desktop Print Client (The Bridge)

The Print Client is a background Node.js daemon running on the print station computer (the PC physically wired to the printer).

Operations Flow:

1. **Startup Setup:** On launch, the client checks if `SumatraPDF.exe` exists locally. If missing, it runs a silent Windows PowerShell command to download the ZIP, extract it, and rename it to `SumatraPDF.exe`.

2. **SSE Event Listening:** The client connects to the server's `/api/jobs/stream`.

3. **Job Detection:** Receives the `new_job` event instantly and polls the server for details.

4. **Download & Print Pipeline:**

- Downloads the document from the server to `print-client/temp`.
- **DOCX to PDF Conversion:** Since SumatraPDF cannot print `.docx` files natively, if the downloaded file is a Word document (`.docx`), the client executes a silent PowerShell script using **Microsoft Word COM Automation** to convert it to a standard `.pdf` before printing:

```
$word = New-Object -ComObject Word.Application
$word.Visible = $false
$doc = $word.Documents.Open("$docPath")
$doc.SaveAs("$pdfPath", 17) # 17 = wdExportFormatPDF
$doc.Close()
$word.Quit()
```

- Spools the resulting `.pdf` file to the printer.

5. **Status Reporting:** Updates status to `printing` (updates progress bar page-by-page) then `completed`, and cleans up all local temp files.

4. How the System Customizes the Print Job (Your Configurations)

The print client maps the student's exact choices to SumatraPDF's CLI engine, translating user-interface selections directly into hardware commands.

A. Print Mode (Black & White vs. Color)

- **Black & White:** The print client sets `-print-settings "monochrome"`. SumatraPDF forces the printer driver to convert colored text and images into high-contrast grayscale.
- **Color:** The print client sets `-print-settings "color"`. This unlocks full-color output on CMYK printers.

B. Page Selection & Ranges

If the student only wants to print specific pages (e.g. pages 1-3 and page 5 out of a 10-page document), the system handles this using the `pages` setting:

- The system takes the page range string (e.g., ``1-3,5``) and appends it to SumatraPDF's settings:

```
SumatraPDF.exe -print-to-default -print-settings "fit,pages=1-3,5"
"document.pdf"
```

- SumatraPDF tells the Windows print spooler to extract and process only those specific pages, leaving the rest of the document untouched.

C. Copies Management

- Instead of spooling the same file multiple times, the client passes `copies=N` directly to SumatraPDF:

```
-print-settings "copies=3"
```

- The printer driver handles collation internally, ensuring the pages come out in order (`Page 1, 2, 3` then `Page 1, 2, 3`... instead of `Page 1, 1, 1` then `Page 2, 2, 2`...).

D. Advanced: Mixed Grayscale and Color (Hybrid Printing)

If a print job requires some pages in color and others in Black & White:

1. **Auto Color Detect:** Most modern ethernet printers automatically print grayscale pages in black toner only (saving color click charges) when the document is printed in `"color"` mode.
2. **Sub-Spooling (Programmatic Split):** If a printer driver does not support auto-detection, the print client can split the print job into two separate commands:
 - *Command 1 (B&W pages)*: `SumatraPDF.exe -print-settings "monochrome,pages=1-2" "file.pdf"`
 - *Command 2 (Color page)*: `SumatraPDF.exe -print-settings "color,pages=3" "file.pdf"`

This sends them as two consecutive jobs to the printer, executing hybrid printing automatically.

5. Security & Robustness Standards

To prevent server crashes, data breaches, or device locks:

1. **Double Sandboxed Network:** The local printing station PC is on a local ethernet subnet, communicating with the server solely via secure outbound HTTPS. No inbound ports are open on the local PC, preventing external attackers from hacking your local network.

2. **Payload Scanning:** Files are scanned for extension validity and byte signatures upon request reception.
 3. **Queue Isolation:** If a print job encounters a print failure, the client tags the job as **failed** in the database and moves on to the next task in the queue, preventing a bad file from blocking subsequent student prints.
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6. Printer Connection & Hardware Spooling

This is the physical layer where the desktop talks to your printer.

How the Computer Sends the File to the Printer:

1. **Silent Spooling:** The print client executes the SumatraPDF command in the background. Because it is sent directly to the printer spooler using silent CLI parameters, **no file dialog, no popup windows, and no saving prompts appear on the desktop screen.**
 2. **Physical Link (Ethernet / USB):**
 - **USB Connection:** Windows sends the spool data packet-by-packet via the USB port driver.
 - **Ethernet Connection:** The spooler sends TCP/IP packets over your local network router directly to the printer's static IP address (typically on port `9100` RAW or `515` LPR).
 3. **Hardware Print:** The printer receives the raw page data, parses it, heats the fuser unit, and prints the pages out physically onto the paper tray.
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